

Shopee Product Matching: Multimodal Finale Report

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1. Approach

Task and data. Group product listings that are the same product. Kaggle "Shopee - Price Match Guarantee", train.csv only: 34,250 listings in 11,014 label groups (median group size 2, maximum 51). Split by group (GroupShuffleSplit 70/15/15, seed 42): validation 5,115 listings / 1,652 groups, test 5,411 / 1,653. Metric: mean per-listing F1 between the predicted and the true set of matches (a listing always matches itself). A system that returns only itself scores 0.4629 (validation) and 0.4401 (test), which is the floor.

Final system: score-level fusion. For each pair of listings, $S = w * S_{\text{text}} + (1 - w) * S_{\text{img}}$. S_{text} is the cosine similarity of word TF-IDF vectors of the titles (vectoriser fitted on the unlabelled titles of the evaluation pool). S_{img} is the cosine similarity of L2-normalised ResNet50 (ImageNet weights, classifier removed, frozen) embeddings. Two listings are matched when S is at least a threshold. The weight $w = 0.45$ and threshold 0.52 were chosen on the validation set by a grid search (w 0.30 to 0.70, threshold 0.35 to 0.65 in steps of 0.01). External resources: torchvision ResNet50 and OpenAI CLIP ViT-B/32 weights and a multilingual MiniLM sentence encoder (used as pretrained feature extractors in experiments), scikit-learn TF-IDF.

2. Experiments

Each experiment follows hypothesis, experiment, result, analysis, conclusion. Validation F1 unless stated.

Experiment	Hypothesis	Result	Conclusion
Text: TF-IDF fitted on train (baseline)	word TF-IDF is a strong text baseline	0.7632 (threshold 0.45)	baseline
Keep single characters	size tokens like 6W/8W matter	0.7650	small gain
Fit TF-IDF on the val/test pool titles	pool vocabulary removes unseen-word gaps	0.7770 (0.43); test 0.7741	kept: final text model
Character n-grams (2 to 4)	handles misspellings and variants	0.7774	no real gain, word TF-IDF kept (simpler)
Multilingual MiniLM embeddings	semantic embeddings beat TF-IDF	0.6424 (0.75)	worse: unrelated-pair mean similarity 0.219 vs 0.004 for TF-IDF
Image: phash	near-duplicate photos	0.6074	weak
Image: frozen ResNet50 / CLIP	deep features beat phash	0.6917 / 0.6616; test (ResNet50) 0.6800	ResNet50 kept
Image: fine-tuned ResNet50 (ArcFace, layer4 + 512-d neck)	metric learning helps on this domain	0.7733; test 0.7743	+0.082 over frozen
Fusion (text + frozen image)	text and image errors are complementary	0.8524 (w 0.45, 0.52); test 0.8508	final system

3. Analysis

Text and image fail in different places, which is why fusion helps. Text fails when the same product is described with different vocabulary (for example model codes such as RT100Q and RT130), and the image fails when different groups share identical photos. Fusion closes about 73% of the gap between the floor and a perfect score on both validation and test (0.4629 to 0.8524 and 0.4401 to 0.8508). Fine-tuning the image model was the largest single image gain (0.6917 to 0.7733). The generic sentence embeddings were worse than TF-IDF here: titles are short, noisy and mix Indonesian and English, and the embedding similarities for unrelated pairs were not close to zero, which makes a single threshold harder. Character n-grams matched word TF-IDF, so the simpler method was kept. Choosing the fusion weight and threshold on validation and reporting test once avoids tuning on test. The final system uses the frozen ResNet50: the fine-tuned embeddings could not be transferred between sessions in time (see limitations).

4. Error analysis (validation, final system)

At threshold 0.52 the system finds 8,315 correct pairs, makes 1,729 false matches and misses 2,696 true pairs (unordered pairs). 900 of the false matches have a high text score only; 2,079 of the misses (77%) have both text and image scores low.

Case	Predicted	Truth	Why it failed	Possible fix
Ecolink 6W vs 8W	same product (fused 1.00)	different groups	identical photo and titles that differ only in a single-character size token, which the tokenizer drops	keep numeric and size tokens, number-aware matching

Case	Predicted	Truth	Why it failed	Possible fix
Bubble-wrap variants	same product	different size groups	same photo reused for several sizes	use the size in the title, group-level reasoning
Salsa Dynamatte	different products (text 0.55 to 0.64, image 0.39, fused below 0.52)	same group	different photos of the same product and loosely matching titles	fine-tuned image model, learned fusion, lower threshold for strong text
Groups with many varied photos (LVN, i7s, Blue Band)	split into pieces	one group	very diverse images, short titles	graph clustering over the pair scores

By group size, validation F1 for text / image / fused is 0.803 / 0.765 / 0.858 for pairs and 0.686 / 0.486 / 0.766 for groups of 11 or more, so large groups are the hardest and the image signal is weakest there.

5. Ablation study

Configuration	Text	Image	Other	Threshold	Val F1	Test F1
A: text only	yes	no	no	0.43	0.7770	0.7741
B: image only (frozen ResNet50)	no	yes	no	0.79	0.6917	0.6800
C: text + image (final)	yes	yes	no	0.52	0.8524	0.8508
D: C + phash, weight 0.1	yes	yes	phash	0.53	0.8525	not run
D: C + phash, weight 0.2	yes	yes	phash	0.53	0.8490	not run
D: C + phash, weight 0.3	yes	yes	phash	0.53	0.8395	not run

Both modalities contribute: adding the image to text gives +0.075 validation F1 over text alone, and adding text to the image gives +0.161. Adding phash does not help (0.8525 at best, then worse), so it is not part of the final system. A centred variant of the embeddings tied with the raw one (0.8494 vs 0.8524), so the simpler raw version was kept.

6. Limitations

- The final fusion uses the frozen ResNet50 (image-only 0.6917), not the fine-tuned model (0.7733), because the fine-tuned embeddings were not available in the fusion session; the final system therefore under-uses the best image model.
- The TF-IDF vectoriser is fitted on the unlabelled titles of the evaluation pool (a transductive but label-free choice); a deployed system would need a fixed vocabulary.
- One split and one seed; no confidence intervals; the threshold and weight are global and not tuned per group size.
- Pairwise thresholding is not transitive and no clustering step is used, so large groups fragment. About 0.5% of image hashes occur in several groups, so some labels are noisy.
- The fusion weight and threshold were tuned on validation only; test was run once for each configuration.

7. Future improvements

- Use the fine-tuned image embeddings in the fusion and re-tune the weight and threshold on validation.
- Number-aware tokenisation (keep size, power and quantity tokens) to separate variants such as 6W and 8W.
- Learned fusion (for example logistic regression or a small network over text similarity, image similarity and phash distance) and a clustering step such as connected components or community detection over the pair scores.
- Hard-negative mining for the metric-learning stage, and OCR of text inside product photos.
- Approximate nearest-neighbour search for scale; multilingual or character-level text encoders.